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* Unit 6: Multiple Integrals & their applications

chapter 9: Double Integration and Area.

* **Text book:-** Higher Engineering Mathematics By B.S. Grewal (Khanna Publications Delhi).

Page Numbers $\rightarrow$ 274 to 302.

* **Reference Book:-** Advanced Engineering Mathematics By Erwin Kreyszig (Wiley Eastern Ltd).

Page No $\rightarrow$ 426 to 483.

* **Applications in Engineering**

- ① One of the major application of multiple integrals in engineering, particularly structure and mechanics, is the determination of properties of plane (2D) and solid bodies (3D). Their volumes, mass, centre of gravity, Moment of Inertia etc. -
- ② To determine Electric charge.
- ③ Computing the work done by a varying force.
- ④ Computing average value of a function.

* **Double Integration:-** Consider the region R is a rectangle which can be denoted as

$$\iint_R f(x,y) dx dy = [a,b] \times [c,d].$$

This means that x and y are
 $a \leq x \leq b$ and $c \leq y \leq d$.

* Direct Evaluation of double integrals.

Ex) The value of double integral $\int_0^1 \int_0^1 \frac{1}{(1+x^2)(1+y^2)} dx dy$

$$\Rightarrow \text{Let } I = \int_0^1 \int_0^1 \frac{1}{(1+x^2)(1+y^2)} dx dy.$$

(Here both of the limits of integration are constant and variables can be separated)

$$\therefore I = \int_0^1 \left(\frac{1}{1+x^2} \right) dx \cdot \int_0^1 \left(\frac{1}{1+y^2} \right) dy.$$

$$I = [\tan^{-1}(x)]_0^1 \cdot [\tan^{-1}(y)]_0^1 \quad \because \int \frac{dx}{1+x^2} = \tan^{-1}(x)$$

$$I = [\tan^{-1}(1) - \tan^{-1}(0)] \cdot [\tan^{-1}(1) - \tan^{-1}(0)]$$

$$I = \left(\frac{\pi}{4} - 0 \right) \left(\frac{\pi}{4} - 0 \right) \quad \left(\because \tan\left(\frac{\pi}{4}\right) = 1 \right)$$

$$\boxed{I = \frac{\pi^2}{16}}$$

② Evaluate $I = \int_0^1 \int_0^1 \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-y^2}} dx dy$.

$$\Rightarrow \text{Let } I = \int_0^1 \int_0^1 \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-y^2}} dx dy$$

$$\therefore I = \int_0^1 \frac{1}{\sqrt{1-x^2}} dx \cdot \int_0^1 \frac{1}{\sqrt{1-y^2}} dy$$

$$I = [\sin^{-1}(x)]_0^1 \cdot [\sin^{-1}(y)]_0^1 \quad \because \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x)$$

$$I = [\sin^{-1}(1) - \sin^{-1}(0)] \cdot [\sin^{-1}(1) - \sin^{-1}(0)]$$

$$I = \left(\frac{\pi}{2} - 0\right) \left(\frac{\pi}{2} - 0\right) = \frac{\pi^2}{4}$$

$$\boxed{I = \frac{\pi^2}{4}}$$

③ Evaluate $\int_0^2 \int_0^y xy \, dx \, dy$

$\Rightarrow$ Let $I = \int_0^2 \int_0^y xy \, dx \, dy$

Here limits of inner integral are in terms of y , therefore integrate w.r.t. x first.

$$I = \int_0^2 \left(\int_0^y x \, dx \right) y \, dy$$

$$I = \int_0^2 \left[\frac{x^2}{2} \right]_0^y y \, dy$$

$$I = \int_0^2 \left(\frac{y^2}{2} - \frac{0^2}{2} \right) y \, dy$$

$$I = \int_0^2 \frac{y \cdot y^2}{2} \, dy = \int_0^2 \frac{y^3}{2} \, dy$$

$$I = \frac{1}{2} \left[\frac{y^4}{4} \right]_0^2$$

$$I = \frac{1}{2} \left[\frac{2^4}{4} - \frac{0^4}{4} \right]$$

$$I = \frac{1}{2} \left[\frac{16}{4} \right] = 2$$

$$\boxed{I = 2}$$

H.W. Evaluate $\int_{-2}^1 \int_{x^2}^{(2-x)} y \, dx \, dy$

* Note! - If limits of inner integral are in terms of x then integrate w.r.t. ' y ' first.

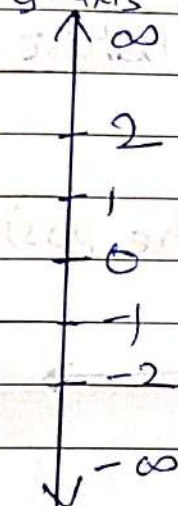
* Double Integration when limits of integration are not given.

(i) Lower limit:- Which gives lowest value of variables x and y .

(ii) Upper limit:- Which gives highest value of variables x and y .



x increases from left to right.

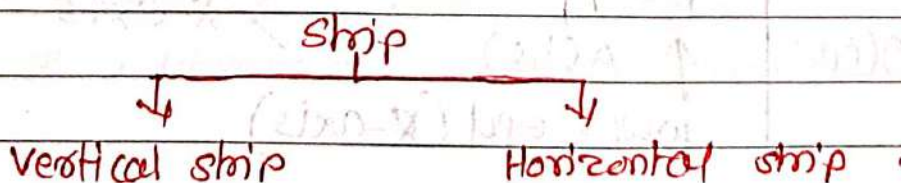


y increases from bottom to top

* Strip:- Strip is the most important tool to find limits of integration.

① To find limits of x , vary the strip from left (gives lower limit) to right (gives upper limit)

② To find limits of y , vary the strip from bottom (gives lower limit) to top (gives upper limit).



* **Vertical ship:** - This kind of ship is always parallel to y-axis. Therefore, we will get first limits of y (represent inner integral) then we will work on limits of x. (which represents on outer integral, which are constant).

* **Horizontal ship:** - This kind of ship is always parallel to x-axis. Therefore, we will get limits of x (represent inner integral) and then we will work on limits of y. (represent on outer integral, which are constant)

* **Note:** - choice of ship depends on nature of the given region.

Que) Evaluate $\iint_R x^2 y^2 dx dy$ over the positive quadrant of the circle $x^2 + y^2 = 1$

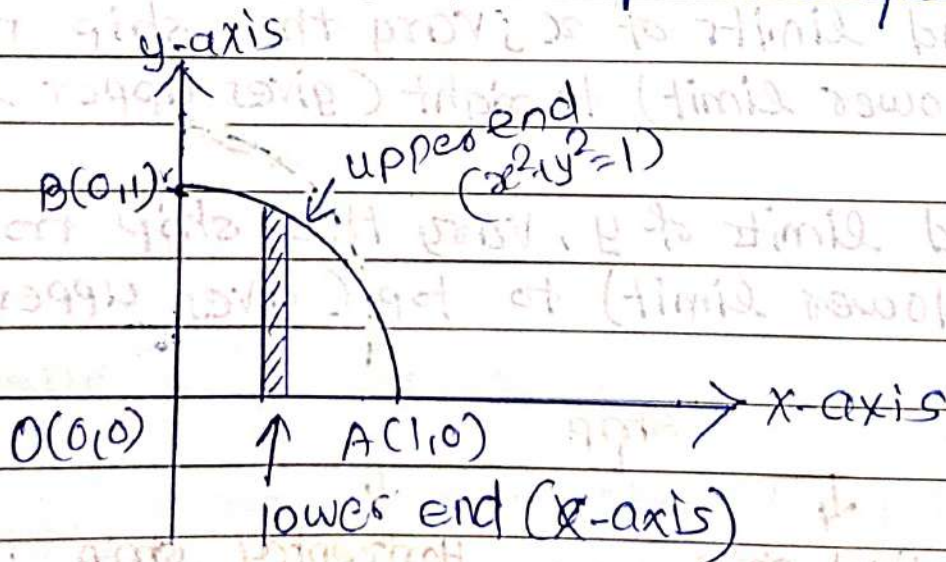
$\Rightarrow$ let $I = \iint_R x^2 y^2 dx dy \longrightarrow (1)$

Step I) diagram

The given eqⁿ of circle is $x^2 + y^2 = 1^2$

$\therefore c \equiv (0,0)$ and radius = 1.

$\therefore$ Diagram of circle in positive quadrant is



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* step ② points of Intersection
Here we already know the points of intersection $O(0,0)$, $A(1,0)$ and $B(0,1)$.

* step ③ ship: -

Let's take vertical ship which is parallel to y -axis.
therefore first we will get limits of y .
i.e. lower limit of y and upper limit of y .

$\therefore$ From fig. Lower end of the ship lies on x -axis
i.e. $y=0$.

and upper end of the ship lies on circle

$$\therefore x^2 + y^2 = 1$$

$$y^2 = 1 - x^2$$

$$y = \pm \sqrt{1 - x^2}$$

for first quadrant y is positive

$$y = \sqrt{1 - x^2}$$

Now to find the limits of x , vary the ship from left to right.

$\therefore$ Taking ship to the left end of curve we get $x=0$

Taking ship to the right end of curve we get $x=1$.

* step ④ Evaluate the limit.

$$I = \int_{x=0}^1 \int_{y=0}^{\sqrt{1-x^2}} x^2 y^2 dx dy$$

$$I = \int_{x=0}^1 \int_{y=0}^{\sqrt{1-x^2}} x^2 y^2 dx dy$$

Here limits of inner integral are in terms of x , therefore integrate w.r.t. y first.

$$I = \int_0^1 \left[\int_0^{\sqrt{1-x^2}} y^2 dy \right] x^2 dx$$

$$I = \int_0^1 \left[\frac{y^3}{3} \right]_0^{\sqrt{1-x^2}} x^2 dx$$

$$I = \int_0^1 \left(\frac{(\sqrt{1-x^2})^3}{3} - \frac{0^3}{3} \right) x^2 dx$$

$$I = \int_0^1 \frac{(1-x^2)^{3/2}}{3} \cdot x^2 dx$$

$$(x^m)^n = x^{mn}$$

put $x^2 = t \Rightarrow x = \sqrt{t} = t^{1/2}$
 $2x dx = dt$

$$\boxed{dx = \frac{dt}{2\sqrt{t}}}$$

since $x=0 \Rightarrow t=0^2=0$
 $x=1 \Rightarrow t=1^2=1$

$\therefore$

$$I = \frac{1}{3} \int_0^1 (1-t)^{3/2} t \cdot \left(\frac{dt}{2\sqrt{t}} \right)$$

$$I = \frac{1}{3} \cdot \frac{1}{2} \int_0^1 (1-t)^{3/2} t^{1-\frac{1}{2}} dt$$

$$I = \frac{1}{6} \int_0^1 (1-t)^{3/2} t^{1/2} dt$$

$m = \frac{3}{2}$ and $n = \frac{1}{2}$ $\int_0^1 (1-x)^m x^n dx = \frac{\beta(m+1, n+1)}{\beta(m+1, n+1)}$

$$\therefore I = \frac{1}{6} \beta\left(\frac{3}{2}+1, \frac{1}{2}+1\right)$$

$$I = \frac{1}{6} \beta\left(\frac{5}{2}, \frac{3}{2}\right)$$

$$I = \frac{1}{6} \cdot \frac{\sqrt{\frac{5}{2}} \cdot \sqrt{\frac{3}{2}}}{\sqrt{\frac{5}{2} + \frac{3}{2}}} \quad \beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$I = \frac{1}{6} \cdot \frac{3/2 \cdot 1/2 \sqrt{\pi} \cdot \frac{1}{2} \sqrt{\pi}}{\sqrt{4}}$$

$$I = \frac{3\pi/8}{6 \times 8!}$$

$$I = (0-10)$$

$$I = \frac{3\pi}{6 \times 8 \times 62} = \frac{\pi}{96}$$

$$\therefore I = \frac{\pi}{96}$$

Que
H.W.

Evaluate $I = \iint \frac{1}{x^4 + y^2} dx dy$ over the region $y \geq x^2$ and $x \geq 1$.

$$\Rightarrow \begin{matrix} y = x^2 \text{ to } y = \infty \\ x = 1 \text{ to } x = \infty \end{matrix} \quad I = \frac{\pi}{4}$$

* Conversion of Cartesian (x, y) double integral to Polar (r, θ) Double integral.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

and

$$x^2 + y^2 = r^2$$

* **Start Ship**:- This ship is used to find r and θ .

If the given integral is related to circle then we use polar system. i.e. $x^2 + y^2 = a^2$.

Que) Evaluate $\iint x^2 y^2 dx dy$ over the positive quadrant of circle $x^2 + y^2 = 1$.

$$\Rightarrow \text{Step 1) Let } I = \iint x^2 y^2 dx dy \longrightarrow \textcircled{1}$$

converting to polar by using

$$x = r \cos \theta, \quad y = r \sin \theta \quad \text{and} \quad dx dy = r dr d\theta$$

$$I = \iint (r \cos \theta)^2 (r \sin \theta)^2 r dr d\theta$$

$$I = \iint r^2 \cos^2 \theta \cdot r^2 \sin^2 \theta \cdot r dr d\theta$$

$$I = \iint r^5 \cos^2 \theta \sin^2 \theta dr d\theta$$

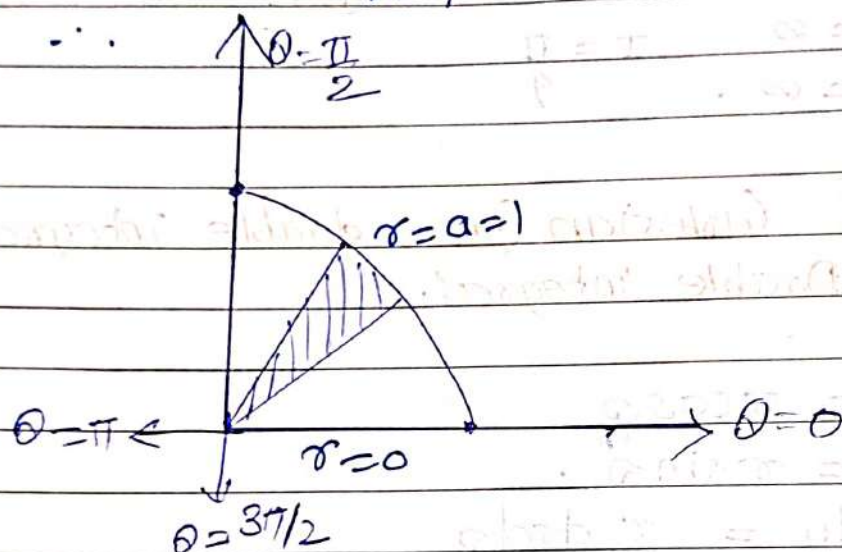
Step ② Diagram

Given $x^2 + y^2 = 1 \Rightarrow$ circle in +ve quadrant

Now converting to polar

$$r^2 = 1$$

$$r = 1$$



Step ③ Points of intersection - Not applicable here.

Step ④ Strip - Slant strip.

from fig, r varies from $r = 0$ to $r = 1$

and θ varies from $\theta = 0$ to $\theta = \pi/2$

Step ⑤ Apply limits and find integral

As
$$I = \int \int r^5 \sin^2 \theta \cos^2 \theta dr d\theta$$

$$I = \int_{\theta=0}^{\theta=\pi/2} \int_{r=0}^{r=1} r^5 \sin^2 \theta \cos^2 \theta dr d\theta$$

$$I = \int_{\theta=0}^{\theta=\pi/2} \sin^2 \theta \cos^2 \theta d\theta \cdot \int_0^1 r^5 dr$$

By reduction formulae.

$$I = \left[\frac{(2-1)(2-1)}{(4)(4-2)} \cdot \frac{\pi}{2} \right] \times \left[\frac{\pi^6}{6} \right]_0^1$$

$$I = \frac{\pi}{16} \cdot \frac{1}{6} = \frac{\pi}{96}$$

$$\therefore \boxed{I = \frac{\pi}{96}}$$

Que) Evaluate $\iint x^2 y^2 dx dy$ over positive quadrant of $x^2 + y^2 = a^2$ using Polar transformations. $\Rightarrow \frac{\pi a^6}{96}$.

Que) Evaluate $\iint_R \frac{x^2 y^2}{x^2 + y^2} dx dy$ where R is annulus between $x^2 + y^2 = 4$, $x^2 + y^2 = 9$.

$$\Rightarrow \text{Step ① Let } I = \iint_R \frac{x^2 y^2}{x^2 + y^2} dx dy.$$

converting to polar by using

$$x = r \cos \theta, y = r \sin \theta \text{ and } x^2 + y^2 = r^2 \text{ and } dx dy = r dr d\theta.$$

$$\therefore I = \iint \frac{(r^2 \cos^2 \theta)(r^2 \sin^2 \theta)}{r^2} r dr d\theta.$$

$$I = \iint r^3 \cos^2 \theta \sin^2 \theta dr d\theta.$$

Step ② Diagram

$$x^2 + y^2 = 4$$

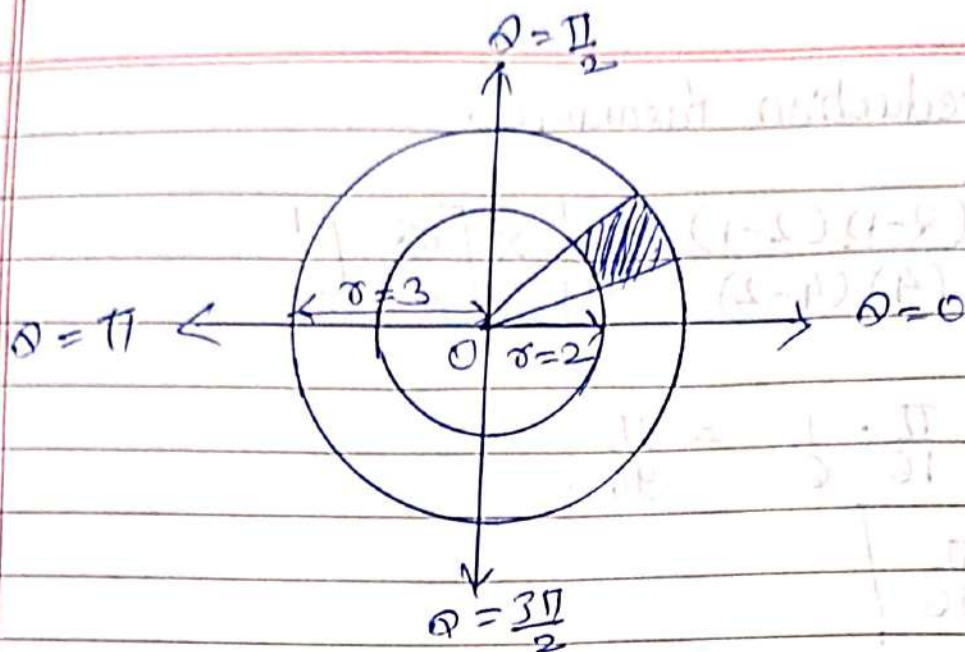
$$r^2 = 4$$

$$r = 2$$

$$x^2 + y^2 = 9$$

$$r^2 = 9$$

$$r = 3$$



step ③ Points of Intersection - NA.

step ④ strip - slant strip

$$r = 2 \text{ to } r = 3.$$

and $\theta = 0$ to $\theta = \pi/2$ (Multiply by 4 to the final answer for symmetry)

step ⑤ find integral

$$I = \iint r^3 \sin^2 \theta \cos^2 \theta \, dr \, d\theta$$

$$I = \int_{\theta=0}^{\theta=\pi/2} \left(\int_{r=2}^{r=3} r^3 \, dr \right) \sin^2 \theta \cos^2 \theta \, d\theta$$

$$\therefore I = \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_2^3 \sin^2 \theta \cos^2 \theta \, d\theta$$

$$I = \int_0^{\pi/2} \left(\frac{3^4}{4} - \frac{2^4}{4} \right) \sin^2 \theta \cos^2 \theta \, d\theta$$

$$I = \left(\frac{81}{4} - \frac{16}{4} \right) \int_0^{\pi/2} \sin^2 \theta \cos^2 \theta \, d\theta$$

By Reduction formula.

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$$I = \frac{65}{4} \left[\frac{(2-1)(2-1)}{4(4-2)} \times \frac{\pi}{2} \right] = \frac{65\pi}{64}$$

By Symmetry

$$I = 4 \times \frac{65\pi}{64} = \boxed{\frac{65\pi}{16}}$$

Que) Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} e^{-x^2-y^2} dx dy$

$\Rightarrow$ $r=0$ to $r=a$, $I = -\frac{\pi}{4} (e^{-a^2}-1)$
 $\theta=0$ to $\theta=\pi/2$

* Area :-

(i) Area of curve in Cartesian (Co-ordinate) System

$$A = \iint dx dy$$

(ii) Area of curve in Polar (Co-ordinate) System.

$$A = \iint r dr d\theta$$

Que) Find the area bounded by the curves $y=x^2$ and $y=x$

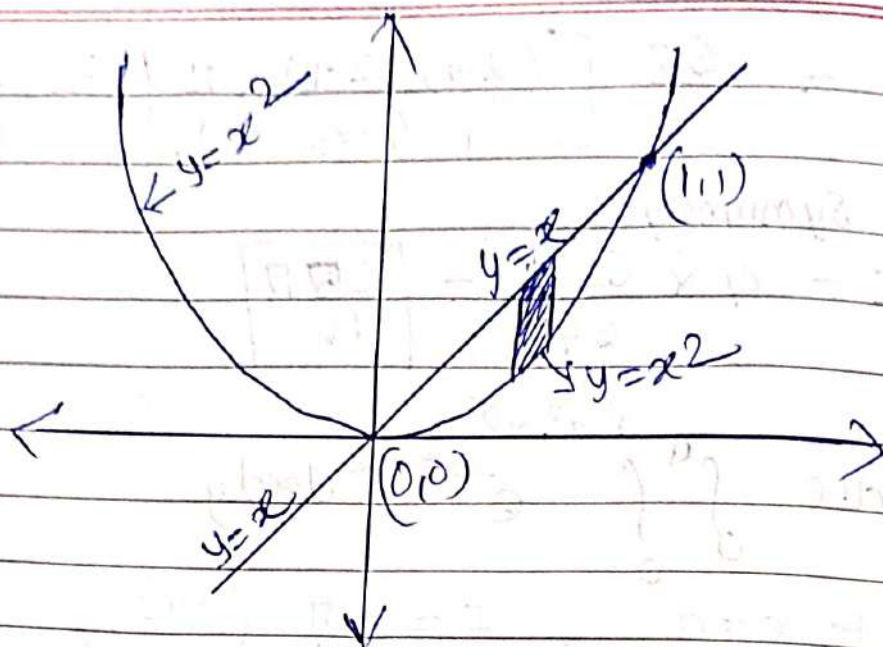
$\Rightarrow$ Step ① Area is given by the formula,

$$A = \iint dx dy$$

Step ② Diagram -

$y=x^2 \rightarrow$ Parabola, Symmetric about Y-axis

$y=x \rightarrow$ line passing through origin
bisecting first quadrant



* Step ③ Points of Intersection

$$y = x^2 \text{ and } y = x$$

$$x = x^2$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$x = 0, x = 1$$

for $x = 0$

$$y = x$$

$$\boxed{y = 0}$$

$(0,0)$ and $(1,1)$.

for $x = 1$

$$y = x$$

$$\boxed{y = 1}$$

* Step ④ ship

Bottom end of the ship lies on parabola.

$$\Rightarrow \therefore y = x^2$$

Top end of the ship lies on line

$$\Rightarrow y = x$$

To find the limits of x , vary the ship from left to right

$\therefore x$ varies from $x = 0$ to $x = 1$.

Step ⑤ Apply the limits.

$$\therefore I = \iint dx dy$$

$$I = \int_0^1 \int_{x^2}^x dx dy$$

$$I = \int_0^1 \left(\int_{x^2}^x dy \right) dx$$

$$I = \int_0^1 [y]_{x^2}^x dx$$

$$I = \int_0^1 (x - x^2) dx$$

$$I = \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1$$

$$I = \left(\frac{1^2}{2} - \frac{1^3}{3} \right) - \left(\frac{0^2}{2} - \frac{0^3}{3} \right)$$

$$I = \frac{1}{6}$$

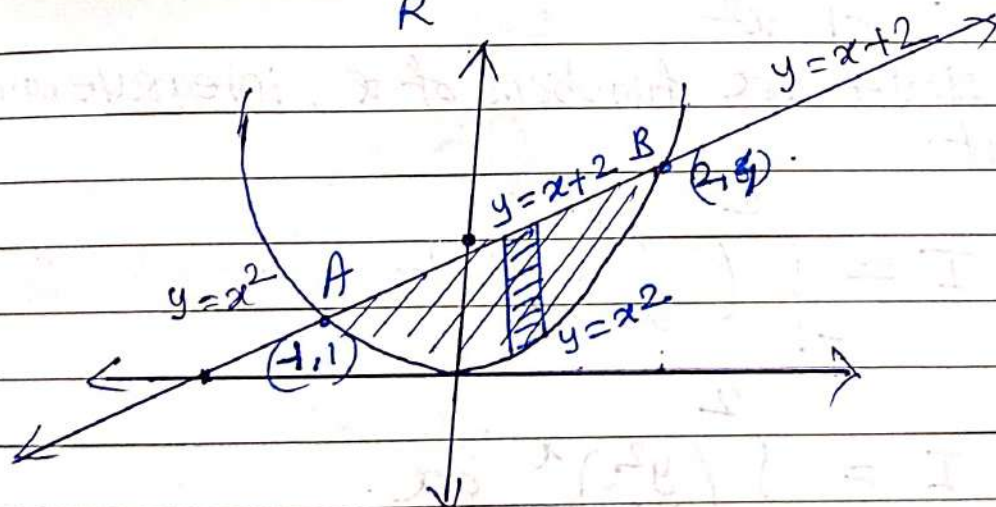
Que) Find the area bounded by $y^2 = 4x$ and $y = x - 8$.

Que) Evaluate $\iint_R y dx dy$ over the region enclosed by the parabola $x^2 = y$ and the line $y = x + 2$

H.W.

$\Rightarrow$ let

$$I = \iint_R y dx dy \quad \longrightarrow \textcircled{1}$$



At point A and B curves intersect to each other

$$y = x^2 \text{ and } y = x + 2$$

$$y = x^2 \Rightarrow x^2 = x + 2$$

$$x^2 - x - 2 = 0$$

$$x^2 + x - 2x - 2 = 0$$

$$x(x+1) - 2(x+1) = 0$$

$$(x+1)(x-2) = 0$$

$$x+1=0, \quad x-2=0$$

$$\boxed{x=-1} \quad \boxed{x=2}$$

$$\text{When } x = -1 \Rightarrow y = x + 2$$

$$y = -1 + 2$$

$$\boxed{y=1} \quad \text{i.e. } (-1, 1)$$

$$\text{When } x = 2 \Rightarrow y = 2 + 2 = 4$$

$$\text{points i.e. } (2, 4)$$

of intersection are $(-1, 1)$ and $(2, 4)$.

In region R strip is parallel to y-axis

$\therefore$ Limits of y are $y = x^2$ and $y = x$

$\therefore$ Limits of x are $x = -1$ to $x = 2$

$\therefore$ eqn ① becomes

$$I = \int_{-1}^2 \int_{x^2}^x y \, dx \, dy$$

inner limits are functions of x, integrate w.r.t. y first.

$$\therefore I = \int_{-1}^2 \left(\int_{x^2}^x y \, dy \right) dx$$

$$I = \int_{-1}^2 \left(\frac{y^2}{2} \right)_{x^2}^x dx$$

$$I = \int_{-1}^2 \left(\frac{x^2}{2} - \left(\frac{x^2}{2} \right)^2 \right) dx$$

$$I = \frac{1}{2} \int_{-1}^2 (x^2 - x^4) dx$$

$$I = \frac{1}{2} \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_{-1}^2$$

$$I = \frac{1}{2} \left[\left(\frac{2^3}{3} - \frac{2^5}{5} \right) - \left(\frac{(-1)^3}{3} - \frac{(-1)^5}{5} \right) \right]$$

$$I = \frac{1}{2} \left[\left(\frac{8}{3} - \frac{32}{5} \right) - \left(-\frac{1}{3} + \frac{1}{5} \right) \right]$$

$$I = \frac{1}{2} \left(\frac{10}{15} \right) = \boxed{\frac{1}{3}}$$

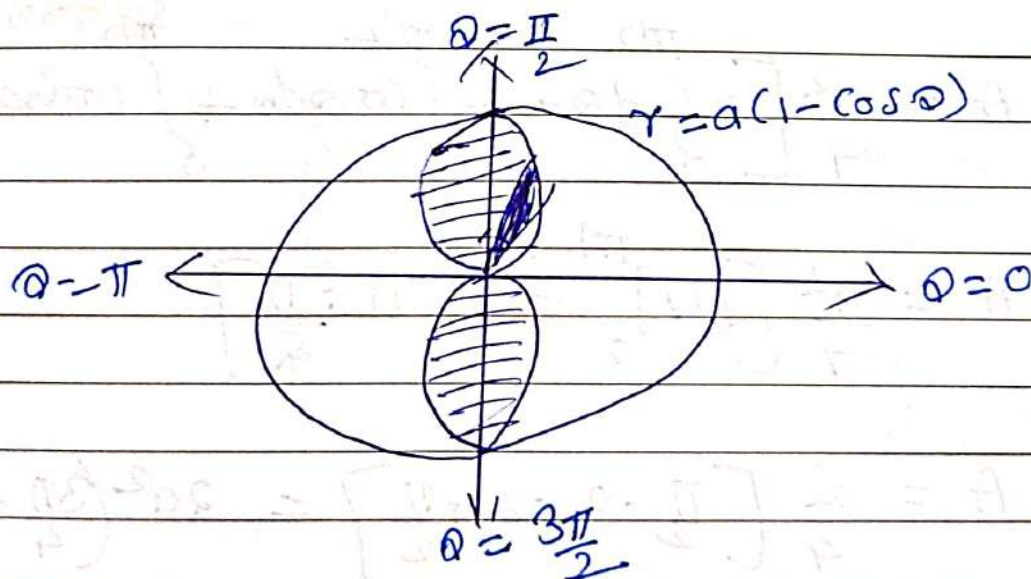
Ques Find the total area included between the cardioid $r = a(1 + \cos \theta)$ and $r = a(1 - \cos \theta)$
 $\Rightarrow$ Area between the curves for polar (r, θ) system is given by

$$A = \iint r dr d\theta$$

Diagram:-

$r = a(1 + \cos \theta) \rightarrow$ Cardioid on $\theta = 0$

$r = a(1 - \cos \theta) \rightarrow$ Cardioid on $\theta = \pi$.



Ship:

$r=0$ to $r=a(1-\cos\theta)$
 $\theta=0$ to $\theta=\frac{\pi}{2}$ (multiply by 4 to final answer for symmetry.

Step 5: Apply the limits.

$$A = \iint r \, dr \, d\theta$$

$$A = \int_0^{\pi/2} \left(\int_0^{a(1-\cos\theta)} r \, dr \right) d\theta$$

$$A = \int_0^{\pi/2} \left(\frac{r^2}{2} \right)_0^{a(1-\cos\theta)} d\theta$$

$$A = \int_0^{\pi/2} \left(\frac{(a(1-\cos\theta))^2}{2} - \frac{0^2}{2} \right) d\theta$$

$$A = \int_0^{\pi/2} \frac{a^2(1-\cos\theta)^2}{4} d\theta$$

$$A = \frac{a^2}{4} \int_0^{\pi/2} (1 - 2\cos\theta + \cos^2\theta) d\theta$$

$$A = \frac{a^2}{4} \left[\theta - 2\sin\theta + \frac{\pi}{4} \right]_0^{\pi/2}$$

$\int_0^{\pi/2} \cos\theta d\theta = 1$
 and
 $\int_0^{\pi/2} \cos^2\theta d\theta = \frac{\pi}{2}$

$$A = \frac{a^2}{4} \left[\int_0^{\pi/2} d\theta - 2 \int_0^{\pi/2} \cos\theta d\theta + \int_0^{\pi/2} \cos^2\theta d\theta \right]$$

$$A = \frac{a^2}{4} \left[\left[\theta \right]_0^{\pi/2} - 2(1) + \frac{\pi}{2} \right]$$

$$A = \frac{a^2}{4} \left[\frac{\pi}{2} - 0 - 2 + \frac{\pi}{2} \right] = 2a^2 \left(\frac{3\pi}{4} - 2 \right)$$

* Change of order of Integration:-

The change of order of integration means if the given ship is vertical, we have to make it horizontal and if the given ship is horizontal, we have to make it vertical.

Que) Change the order of integration in the double integration $\int_0^5 \int_{2-x}^{2+x} f(x,y) dy dx$.

step ①
 $\Rightarrow$ let $I = \int_0^5 \int_{2-x}^{2+x} f(x,y) dy dx$.

Now the limits of inner integral are in terms of x .

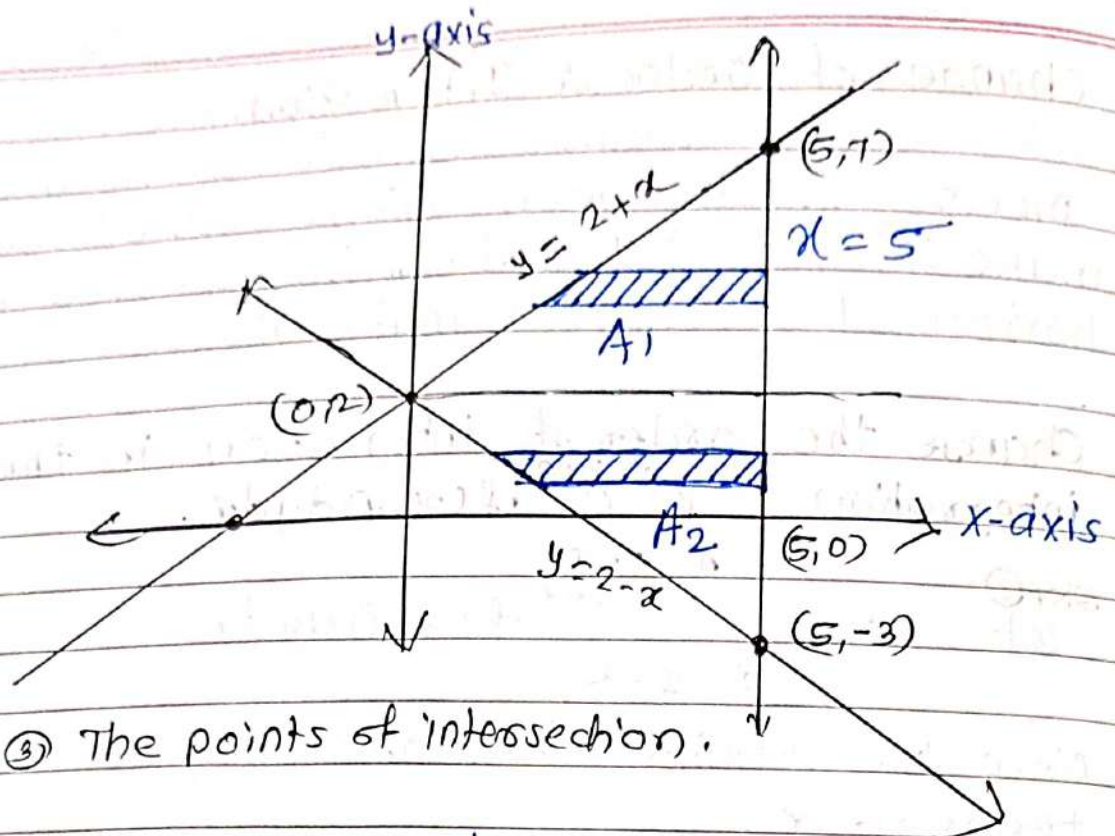
Therefore $y = \text{lower limit}$ and $y = \text{upper limit}$.

As vertical ship is used to find limits and we have to change it to horizontal ship.

$$I = \int_{x=0}^{x=5} \int_{y=2-x}^{y=2+x} f(x,y) dx dy. \longrightarrow \textcircled{1}$$

step ② Diagram.

- from ① $\Rightarrow$
- (i) $x = 0 \longrightarrow y\text{-axis}$
 - (ii) $x = 5 \longrightarrow \text{line parallel to } y\text{-axis}$
 - (iii) $y = 2+x \longrightarrow \text{straight line passing through } (0,2) \text{ and } (2,0)$.
 - (iv) $y = 2-x \longrightarrow \text{straight line passing through } (0,2) \text{ and } (2,0)$.



step ③ The points of intersection.

(i) for $x = 5$ and $y = 2 + x$

$$y = 2 + 5$$

$$\boxed{y = 7}$$

i.e. $(5, 7)$

(ii) for $x = 5$ and $y = 2 - x$

$$y = 2 - 5$$

$$\boxed{y = -3}$$

i.e. $(5, -3)$.

* Step ④ Ship :- Horizontal ship

(i) A1 Left end of the ship lies on $y = 2 + x$

$$\therefore x = y - 2$$

Right end of the ship lies on

$$x = 5$$

For the limits of y , vary the ship from bottom to top.

$\therefore y$ varies from

$$y = 2 \text{ to } y = 7.$$

(ii) area A_2 .

Left end of the ship lies on $y = 2 - x$.

$$\therefore [x = 2 - y]$$

$\therefore$ Right end of the ship lies on

$$[x = 5]$$

y varies from $y = -3$ to $y = 2$.

* step ⑤
$$I = \int_0^5 \int_{2-x}^{2+x} f(x, y) dx dy$$

$$\therefore I = \int_2^7 \int_{y-2}^5 f(x, y) dy dx + \int_{-3}^2 \int_{2-y}^5 f(x, y) dx dy$$

Ex)
$$\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dx dy$$

$\Rightarrow$ step ① let
$$I = \int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dx dy$$

For the change of order of integration.

$$I = \int_{x=0}^{x=\infty} \int_{y=x}^{y=\infty} \frac{e^{-y}}{y} dy dx \longrightarrow \textcircled{1}$$

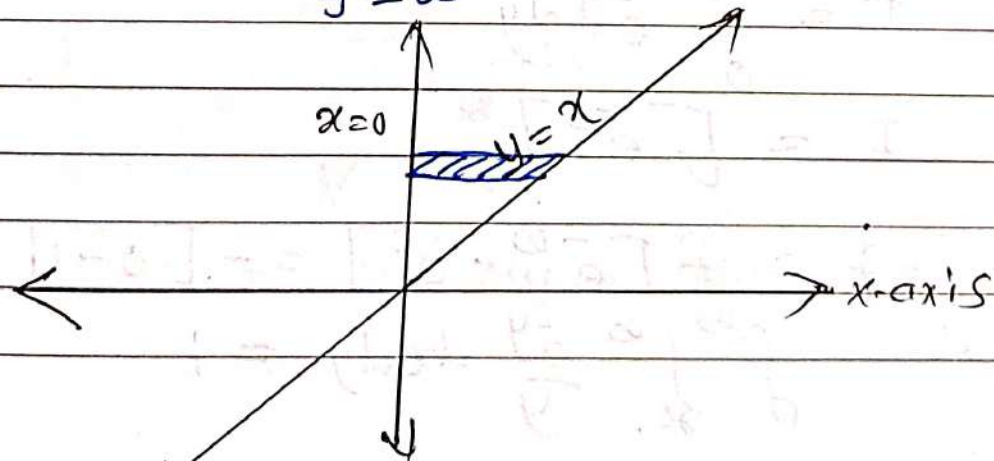
* step ② Diagram:-

$\therefore$ from ① $x = 0 \rightarrow Y\text{-axis}$

$$x = \infty$$

$y = x \rightarrow$ Line bisecting 1st quadrant.

$$y = \infty$$



step ③ $\therefore$ Point of intersection.

step ④ Ship.

In given integral the inner limits are in terms of x .

$$\therefore y = x \text{ and } y = \infty.$$

Now ~~these~~ used vertical ship.

$\therefore$ Now convert to horizontal ship.

Left end of the ship lies on

$$x = 0$$

Right end of the ship lies on

$$x = y$$

$\therefore y$ varies from bottom to top

$$\therefore y = 0 \text{ to } y = \infty.$$

step ⑤ Now Apply new limits

$$\therefore I = \int_{y=0}^{y=\infty} \int_{x=0}^{x=y} \left(\frac{e^{-y}}{y} \right) dy dx.$$

$$I = \int_0^{\infty} \left(\int_0^y dx \right) \frac{e^{-y}}{y} dy$$

$$I = \int_0^{\infty} [x]_0^y \cdot \frac{e^{-y}}{y} dy$$

$$I = \int_0^{\infty} (y-0) \frac{e^{-y}}{y} dy$$

$$I = \int_0^{\infty} e^{-y} dy$$

$$I = [e^{-y}]_0^{\infty}$$

$$I = -[e^{-\infty} - e^0] = -[0 - 1] = 1$$

$$\therefore \int_0^{\infty} \int_0^{\infty} \frac{e^{-y}}{y} dx dy = 1.$$

Que
H.W.

find $\int_0^{\infty} \int_y^{\infty} \frac{e^{-x}}{x} dx dy.$

Que) change the order of integration $\int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y} dx dy.$

Que) Evaluate $\int_1^{\infty} dx \int_1^{\infty} e^{-y} y^x \log y dy.$

$\Rightarrow$ let $I = \int_1^{\infty} dx \int_1^{\infty} e^{-y} y^x \log y dy.$

Here it is simpler to integrate first w.r.t. x .
Now as limits of both integrals are constant
we can interchange the order, without using diagram

$$I = \int_1^{\infty} e^{-y} \log y dy \int_1^{\infty} y^x dx$$

$$I = \int_1^{\infty} e^{-y} \log y dy \left[\frac{y^x}{\log y} \right]_1^{\infty} \quad \int a^x dx = \frac{a^x}{\log a}.$$

$$I = \int_1^{\infty} e^{-y} \log y \left[\frac{y^{\infty}}{\log y} - \frac{y^0}{\log y} \right] dy$$

$$I = \int_1^{\infty} e^{-y} \cancel{\log y} \left(\frac{y-1}{\cancel{\log y}} \right) dy$$

$$I = \int_1^{\infty} e^{-y} (y-1) dy$$

$$\int uv = u \int v dy - \int \left(\frac{du}{dy} \int v dy \right) dy$$

$$I = (y-1) \int_1^{\infty} e^{-y} dy - \int_1^{\infty} \left(\frac{d}{dy} (y-1) \int_1^{\infty} e^{-y} dy \right) dy$$

$$I = \left[(y-1) \frac{e^{-y}}{-1} \right]_1^{\infty} - \int_1^{\infty} (1-0) \frac{e^{-y}}{-1} dy$$

$$I = - \left[(\infty-1)e^{-\infty} - (1-1)e^{-1} \right] + \int_1^{\infty} e^{-y} dy$$

$$I = (0-0) + \left[\frac{e^{-y}}{-1} \right]_1^{\infty} \quad (\because e^{-\infty} = 0)$$

$$I = - \left[e^{-\infty} - e^{-1} \right]$$

$$I = - \left[0 - \frac{1}{e} \right]$$

$$\boxed{I = \frac{1}{e}}$$

Unit 6 , chapter - 10

Triple integration and volume.

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* Type ① Direct Evaluation of Triple Integrals.

IF the limits of inner integral are in terms of x and y then integrate w.r.t. z first. Now it is converted to double integral.

* Type ② Triple Integration when limits are not provided.

* **Method 1:-** Eliminate z , That is integrate w.r.t. z by using given limits of z . Then use the same process as used in double integration.

* **Method 2:-** Convert the given integral into polar system for triple integration there are three polar systems.

(i) Spherical Polar System:-

For sphere $x^2 + y^2 + z^2 = a^2$

where $a =$ radius of the sphere

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\therefore x^2 + y^2 + z^2 = r^2$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

* Limits of r , θ and ϕ .

complete sphere	Hemisphere	Positive octant
$r = 0$ to $r = a$	$r = 0$ to $r = a$	$r = 0$ to a
$\theta = 0$ to $\theta = \pi$	$\theta = 0$ to $\theta = \pi/2$	$\theta = 0$ to $\pi/2$
$\phi = 0$ to $\phi = 2\pi$	$\phi = 0$ to $\phi = 2\pi$	$\phi = 0$ to $\pi/2$

(ii)

For Ellipsoide (Ellipse in 3D)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$\text{put } x = ar \sin \theta \cos \phi$$

$$y = br \sin \theta \sin \phi$$

$$z = cr \cos \theta$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = r^2$$

$$dx dy dz = abc r^2 \sin \theta dr d\theta d\phi$$

and limits are

$$r = 0 \text{ to } a$$

$$\theta = 0 \text{ to } \pi$$

$$\phi = 0 \text{ to } 2\pi$$

(iii)

Cylindrical Polar System:-

For cylindrical Polar System (i.e. for cylinder or cone) first integrate w.r.t. z and convert the triple integral to $\iint f(x, y) dx dy$.

then convert this double integral to polar form by using

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$dx dy = r dr d\theta$$

limits are $r = 0 \text{ to } a$

$$\theta = 0 \text{ to } 2\pi$$

*

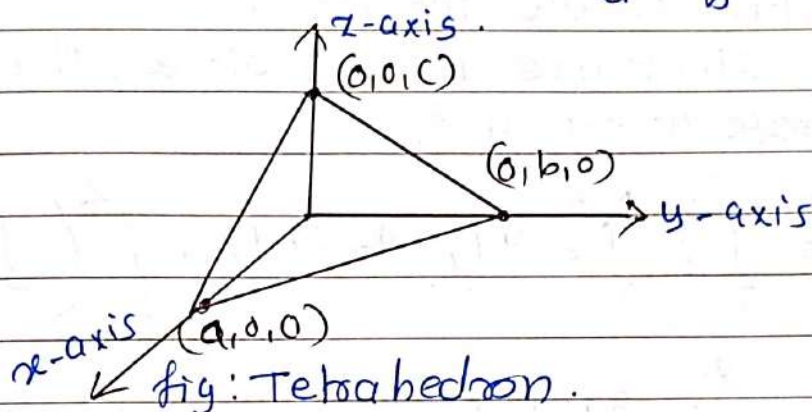
Dirichlet's Theorem:- It is used only when we have to triple integrate over a tetrahedron

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

According to Dirichlet's theorem,

$$\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\Gamma l \cdot \Gamma m \cdot \Gamma n}{\Gamma(l+m+n)}$$

where $x, y, z \geq 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} \leq 1$.



Ex) Evaluate $\int_0^2 \int_0^x \int_0^{2x+2y} e^{x+y+z} dz dy dx$.

$\Rightarrow$ Let $I = \int_0^2 \int_0^x \int_0^{2x+2y} e^x \cdot e^y \cdot e^z dz dy dx$

($\because a^{m+n} = a^m \cdot a^n$)

Here limits of inner integral are in terms of x and y . Therefore integrate w.r.t. z first.

$$I = \int_0^2 \int_0^x \left[\int_0^{2x+2y} e^z dz \right] e^x \cdot e^y dx dy$$

$$I = \int_0^2 \int_0^x \left[e^z \right]_0^{2x+2y} e^x \cdot e^y dx dy$$

$$I = \int_0^2 \int_0^x \left[e^{2x+2y} - e^0 \right] e^x \cdot e^y dx dy$$

($\because e^0 = 1$)

$$I = \int_0^2 \int_0^x \left[e^{2x} \cdot e^{2y} \cdot e^x \cdot e^y - e^x \cdot e^y \right] dx dy$$

($a^{m+n} = a^m \cdot a^n$)

$$I = \int_0^2 \int_0^x [e^{3x} \cdot e^{3y} - e^x \cdot e^y] dx dy$$

$$I = \int_0^2 \int_0^x e^{3x} e^{3y} dx dy - \int_0^2 \int_0^x e^x \cdot e^y dx dy.$$

Now it is double integral.

Inner limits are in terms of x , Therefore integrate w.r.t. y first.

$$I = \int_0^2 \left(\int_0^x e^{3y} dy \right) e^{3x} dx - \int_0^2 \left(\int_0^x e^y dy \right) e^x dx.$$

$$I = \int_0^2 \left[\frac{e^{3y}}{3} \right]_0^x e^{3x} dx - \int_0^2 \left[e^y \right]_0^x e^x dx$$

$$I = \int_0^2 \left(\frac{e^{3x}}{3} - \frac{e^0}{3} \right) e^{3x} dx - \int_0^2 (e^x - e^0) e^x dx$$

$$I = \int_0^2 \left(\frac{e^{3x}}{3} - \frac{1}{3} \right) e^{3x} dx - \int_0^2 (e^x - 1) e^x dx$$

$$I = \frac{1}{3} \int_0^2 e^{3x} \cdot e^{3x} dx - \frac{1}{3} \int_0^2 e^{3x} dx - \int_0^2 e^x \cdot e^x dx + \int_0^2 e^x dx$$

$$I = \frac{1}{3} \int_0^2 e^{6x} dx - \frac{1}{3} \int_0^2 e^{3x} dx - \int_0^2 e^{2x} dx + \int_0^2 e^x dx$$

$$I = \frac{1}{3} \left[\frac{e^{6x}}{6} \right]_0^2 - \frac{1}{3} \left[\frac{e^{3x}}{3} \right]_0^2 - \left[\frac{e^{2x}}{2} \right]_0^2 + \left[e^x \right]_0^2$$

$$I = \frac{1}{18} (e^{12} - e^0) - \frac{1}{9} (e^6 - e^0) - \frac{1}{2} (e^4 - e^0) + (e^2 - e^0)$$

$$(\because e^0 = 1)$$

$$-\frac{1}{18} + \frac{1}{9} + \frac{1}{2} - 1 = \frac{-1 + 2 + 9 - 18}{18} = \frac{-8}{18} = -\frac{4}{9}$$

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$$I = \frac{1}{18} (e^{12} - 1) - \frac{1}{9} (e^6 - 1) - \frac{1}{2} (e^4 - 1) + e^2 - 1$$

$$I = \frac{e^{12}}{18} - \frac{e^6}{9} - \frac{e^4}{2} + e^2 - \frac{1}{18} + \frac{1}{9} + \frac{1}{2} - 1$$

$$I = \frac{1}{18} e^{12} - \frac{1}{9} e^6 - \frac{1}{2} e^4 + e^2 - \frac{4}{9}$$

Que) Evaluate $\iiint \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$ taken throughout the volume of the sphere $x^2+y^2+z^2=1$ in the positive octant.

$\Rightarrow$

$$\text{Let } I = \iiint \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}} \longrightarrow \textcircled{1}$$

Transforming to spherical polar system.

For sphere $x^2+y^2+z^2=a^2$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\text{and } x^2+y^2+z^2 = r^2$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

Limits of r, θ and ϕ for positive octant

$$r = 0 \text{ to } a$$

$$\theta = 0 \text{ to } \pi/2$$

$$\phi = 0 \text{ to } \pi/2$$

Here all the limits are constants and variables can be separated.

$$\Rightarrow I = \int_{\phi=0}^{\pi/2} \int_{\theta=0}^{\pi/2} \int_{r=0}^1 \frac{r^2 \sin \theta dr d\theta d\phi}{\sqrt{1-r^2}}$$

$$I = \int_{\phi=0}^{\pi/2} \int_{\theta=0}^{\pi/2} \sin \theta d\theta \int_{r=0}^1 \frac{r^2}{\sqrt{1-r^2}} dr \rightarrow \textcircled{2}$$

$$\text{let } I_1 = \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr.$$

$$I = \int_{\phi=0}^{\pi/2} d\phi \left(\int_{\theta=0}^{\pi/2} \sin \theta d\theta \right) I_1$$

$$I = \int_0^{\pi/2} d\phi [-\cos \theta]_0^{\pi/2} I_1$$

$$I = - \int_0^{\pi/2} d\phi [\cos \pi/2 - \cos 0] I_1$$

$$I = - \int_0^{\pi/2} d\phi (0 - 1) I_1$$

$$I = + \int_0^{\pi/2} d\phi I_1$$

$$I = [\phi]_0^{\pi/2} I_1$$

$$I = \left[\frac{\pi}{2} - 0 \right] I_1 = \left[\frac{\pi}{2} I_1 \right] \rightarrow \textcircled{*}$$

$$\text{let } I_1 = \int_0^1 \frac{r^2}{\sqrt{1-r^2}} dr$$

$$\text{put } r^2 = t \Rightarrow r = \sqrt{t}$$

$$2r dr = dt \Rightarrow \frac{dt}{2} = r dr$$

$$\text{since } r=0 \Rightarrow t=0^2=0$$

$$r=1 \Rightarrow t=1^2=1$$

$$\therefore I_1 = \int_0^1 \frac{\sqrt{t} \cdot \left(\frac{dt}{2}\right)}{\sqrt{1-t}}$$

$$I_1 = \frac{1}{2} \int_0^1 (1-t)^{-1/2} t^{1/2} dt.$$

$$I_1 = \frac{1}{2} \int_0^1 t^{1/2} (1-t)^{-1/2} dt.$$

By definition of Beta function,

$$I_1 = \frac{1}{2} \beta\left(\frac{1}{2}+1, \frac{1}{2}+1\right)$$

$$I_1 = \frac{1}{2} \beta\left(\frac{3}{2}, \frac{1}{2}\right)$$

$$I_1 = \frac{1}{2} \frac{\Gamma_{3/2} \cdot \Gamma_{1/2}}{\Gamma_{\frac{3}{2} + \frac{1}{2}}}$$

$$I_1 = \frac{1}{2} \cdot \frac{\frac{1}{2} \sqrt{\pi} \cdot \sqrt{\pi}}{\sqrt{2}} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{\pi}{(2-1)!} = \frac{\pi}{4}$$

$\therefore$ eqn (*) becomes

$$I = \frac{\pi}{2} \left(\frac{\pi}{4}\right) = \frac{\pi^2}{8} \rightarrow \text{which is required answer.}$$

Que) Evaluate $\iiint x^2 y z \, dx dy dz$ throughout the volume bounded by plane $x=0, y=0, z=0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\Rightarrow \text{Let } I = \iiint x^2 y z \, dx dy dz. \rightarrow \textcircled{1}$$

$$\text{put } \frac{x}{a} = u, \frac{y}{b} = v, \frac{z}{c} = w.$$

$$\therefore x = au, y = bv, z = cw.$$

Differentiating

$$dx = a du, dy = b dv, dz = c dw.$$

from eqn ①

$$I = \iiint (au)^2 (bv) (wc) (adu) (bdv) (cdw)$$

$$I = a^3 b^2 c^2 \iiint u^2 v w \, du \, dv \, dw$$

where $u+v+w \leq 1$.

$$I = a^3 b^2 c^2 \iiint u^{3-1} v^{2-1} w^{2-1} \, du \, dv \, dw$$

By Dirichlet's theorem. $l=3, m=2, n=2$

$$I = a^3 b^2 c^2 \cdot \left(\frac{\sqrt{3} \sqrt{2} \sqrt{2}}{1 \cdot 3 + 2 + 2 + 1} \right)$$

$$I = a^3 b^2 c^2 \frac{(2!)(1!)(1!)}{7!} \quad \Gamma n = (n-1)!$$

$$I = a^3 b^2 c^2 \left(\frac{2}{7!} \right)$$

$$I = \frac{a^3 b^2 c^2}{2520}$$

H.W. Que) Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) \, dx \, dy \, dz$

⇒ Hint! Separate the integrals.

Inner integrals are in terms of x and z .

therefore first integrate w.r.t. y .

Ans: $I = 0$.

Que) Evaluate $\iiint (x^2 y^2 + y^2 z^2 + z^2 x^2) \, dx \, dy \, dz$.

throughout the volume of the sphere $x^2 + y^2 + z^2 = a^2$
spherical polar system,

$$\Rightarrow I = \frac{4\pi a^7}{35}$$

* Volume *

* Application of Triple Integration $\Rightarrow$ Volume
The volume of solids by using triple integration is given by,

$$V = \iiint dx dy dz.$$

first find the limits of z then integrate w.r.t. z .

$$\iint \left(\int_{z_1}^{z_2} dz \right) dx dy = \iint (z_2 - z_1) dx dy.$$

then solve by using the process of double integration.

Que) Prove that the volume bounded by the cylinders $y^2 = x$ and $x^2 = y$ and the planes $z = 0$ is $x + y + z = 2$ is $\frac{11}{30}$.

$\Rightarrow$ step ① let $V = \iiint dx dy dz$.

but limits of z are $z = 0$ and $z = 2 - x - y$.
(-- given).

$$\therefore V = \iint \left(\int_{z=0}^{z=2-x-y} dz \right) dx dy$$

$$V = \iint [z]_0^{2-x-y} dx dy$$

$$V = \iint (2 - x - y - 0) dx dy.$$

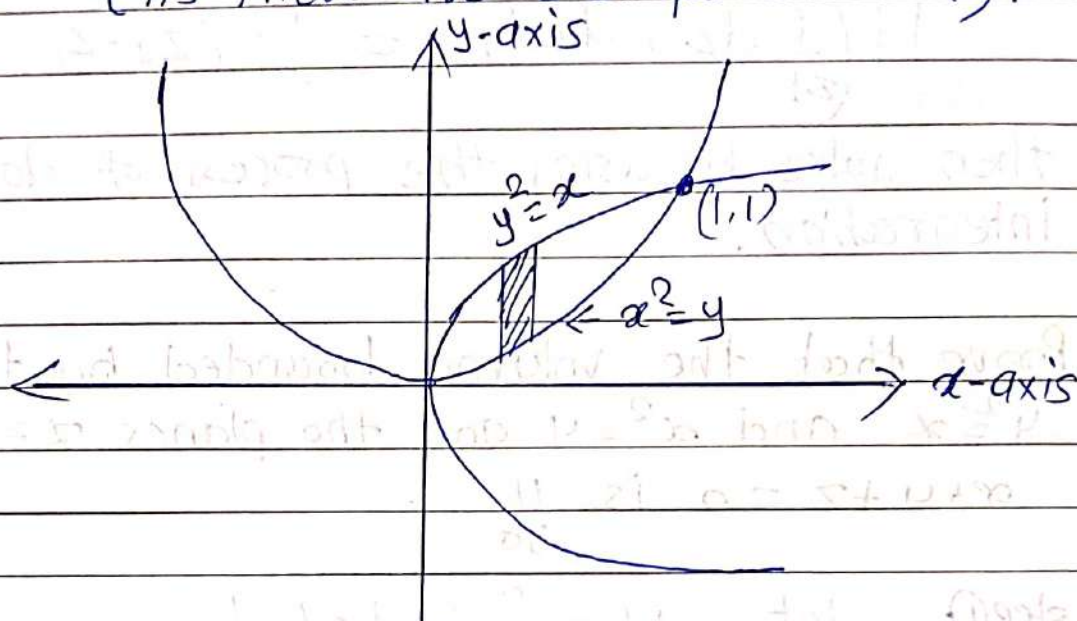
$$V = \iint 2 dx dy - \iint x dx dy - \iint y dx dy. \rightarrow \textcircled{1}$$

$$V = I_1 - I_2 - I_3$$

* Step ② :- Diagram
 Given $y^2 = x \rightarrow$ Parabola,
 Symmetrical about x -axis
 (because even powers of y).

and

$x^2 = y \rightarrow$ Parabola
 Symmetrical about y -axis
 (As there are even powers of x).



* Step ③ Points of Intersection:-

$$\begin{aligned}
 & y^2 = x \text{ and } x^2 = y \\
 & \Rightarrow (y^2)^2 = y \\
 & \Rightarrow y^4 = y \\
 & \Rightarrow y^4 - y = 0 \\
 & \Rightarrow y(y^3 - 1) = 0 \\
 & \Rightarrow y = 0 \text{ or } y^3 = 1 \\
 & \Rightarrow \boxed{y=0} \quad \boxed{y=1}
 \end{aligned}$$

When $y = 0 \Rightarrow x = 0^2 = \boxed{0}$ i.e. $(0,0)$

When $y = 1 \Rightarrow x = 1^2 = 1$ i.e. $(1,1)$

* Step ④ Strip:-

Vertical strip

Bottom end of the ship lies on $y = x^2$

Top end of the ship lies on $y^2 = x$.

$$\Rightarrow y = \pm \sqrt{x}$$

$$y = \sqrt{x} \text{ (in first quad.)}$$

x varies from $x=0$ to $x=1$.

Equation ① becomes

$$V = 2 \int_0^1 \int_{x^2}^{\sqrt{x}} dx dy - \int_0^1 \int_{x^2}^{\sqrt{x}} x dx dy - \int_0^1 \int_{x^2}^{\sqrt{x}} y dx dy$$

Inner limits are in terms of x

~~∴~~ therefore integrate w.r.t. y first.

$$\therefore V = 2 \int_0^1 \left(\int_{x^2}^{\sqrt{x}} dy \right) dx - \int_0^1 \left(\int_{x^2}^{\sqrt{x}} dy \right) x dx - \int_0^1 \left(\int_{x^2}^{\sqrt{x}} y dy \right) dx$$

$$V = 2 \int_0^1 [y]_{x^2}^{\sqrt{x}} dx - \int_0^1 [y]_{x^2}^{\sqrt{x}} x dx - \int_0^1 \left[\frac{y^2}{2} \right]_{x^2}^{\sqrt{x}} dx$$

$$V = 2 \int_0^1 (\sqrt{x} - x^2) dx - \int_0^1 (\sqrt{x} - x^2) x dx - \int_0^1 \frac{1}{2} [(\sqrt{x})^2 - (x^2)^2] dx$$

$$V = 2 \int_0^1 (x^{1/2} - x^2) dx - \int_0^1 (x^{3/2} - x^3) dx - \frac{1}{2} \int_0^1 (x - x^4) dx$$

$$V = 2 \left[\frac{x^{3/2}}{3/2} - \frac{x^3}{3} \right]_0^1 - \left[\frac{x^{5/2}}{5/2} - \frac{x^4}{4} \right]_0^1 - \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1$$

$$V = \left[\frac{4}{3} (1^{3/2}) - \frac{2}{3} (1)^3 - 0 \right] - \left[\frac{2}{5} (1)^{5/2} - \frac{1}{4} - 0 \right] - \frac{1}{2} \left[\frac{1^2}{2} - \frac{1^5}{5} - 0 \right]$$

$$V = \left(\frac{4}{3} - \frac{2}{3} \right) - \frac{2}{5} - \frac{1}{4} - \frac{1}{4} + \frac{1}{10} = \frac{2}{3} - \frac{3}{20} - \frac{3}{20} = \boxed{\frac{11}{30}}$$

$$\therefore \boxed{V = \frac{11}{30}} \text{ Hence proved.}$$

Ques) Find the volume of paraboloid of revolution
 $x^2 + y^2 = 4z$ cut off by the plane $z=4$

⇒ Step ①: - $V = \iiint dx dy dz.$

⇒ The limits of z .

Given $x^2 + y^2 = 4z$ and $z=4$

$$\boxed{z = \frac{x^2 + y^2}{4}}$$

$$\text{and } \boxed{z=4}$$

$$\therefore V = \iint \left(\int_{z=\frac{x^2+y^2}{4}}^{z=4} dz \right) dx dy$$

$$V = \iint \left[z \right]_{\left(\frac{x^2+y^2}{4} \right)}^4 dx dy$$

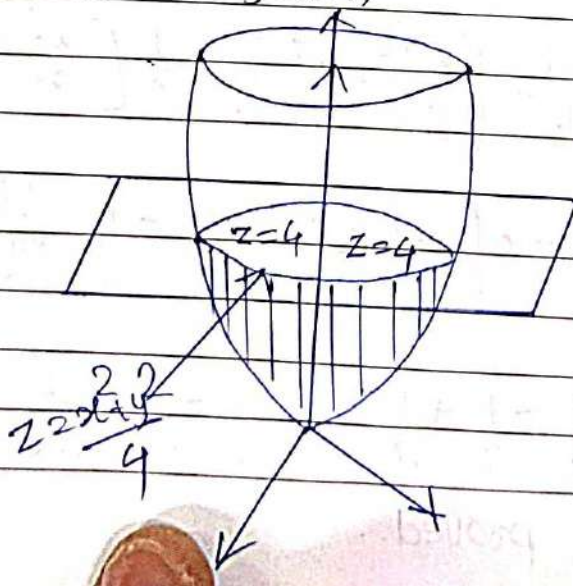
$$V = \iint \left(4 - \frac{x^2+y^2}{4} \right) dx dy$$

By converting into polar

$$V = \iint \left(4 - \frac{r^2}{4} \right) r dr d\theta.$$

$$V = \iint \left(4r - \frac{r^3}{4} \right) dr d\theta.$$

* Step ②: Diagram

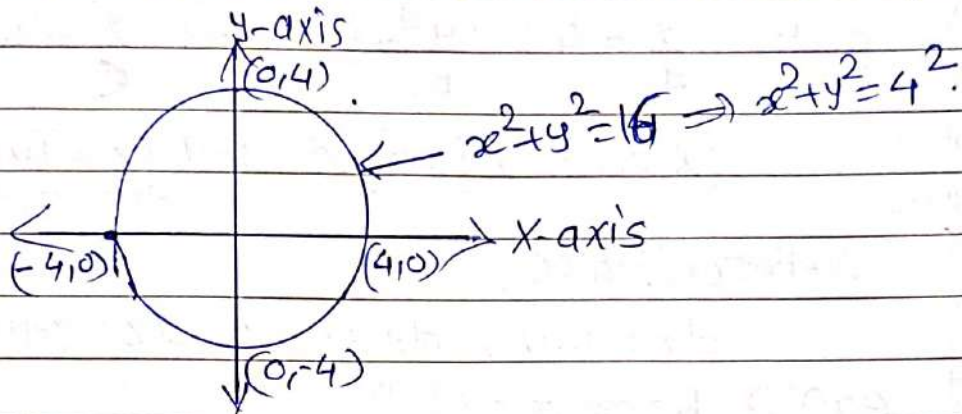


* Step ③ Points of Intersection

Here the paraboloid $x^2 + y^2 = 4z$ is cut off by the plane $z = 4$

$$\therefore x^2 + y^2 = 4(4)$$

$$x^2 + y^2 = 16 \rightarrow \text{circle of radius 4.}$$



* Step ④ Slip.

$$r = 0 \text{ to } r = 4$$

$$\theta = 0 \text{ to } \theta = \pi/4 \quad (\text{multiply by 4 to final answer for symmetry})$$

* Step ⑤

Apply limits from step 4 to step 1.

$$V = \iint \left(4r - \frac{r^3}{4}\right) dr d\theta$$

$$V = \int_0^{\pi/2} d\theta \int_0^4 \left(4r - \frac{r^3}{4}\right) dr$$

$$V = \left[\theta\right]_0^{\pi/2} \left[\frac{4r^2}{2} - \frac{r^4}{16}\right]_0^4$$

$$V = \left(\frac{\pi}{2} - 0\right) \cdot \left[\left(\frac{4(4)^2}{2} - \frac{4^4}{16}\right) - (0 - 0)\right]$$

$$V = \frac{\pi}{2} \cdot [32 - 16]$$

$$V = \frac{16\pi}{2} = 8\pi$$

For Symmetry $V = 4 \times 8\pi = \boxed{32\pi}$

Que) Find the volume of the tetrahedron bounded by co-ordinate plane and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

⇒

$$V = \iiint dx dy dz \quad \longrightarrow \quad \textcircled{1}$$

For tetrahedron we use dirichlet's theorem.

put $\frac{x}{a} = u$, $\frac{y}{b} = v$ and $\frac{z}{c} = w$

$$\boxed{x = au}, \quad \boxed{y = bv}, \quad \text{and} \quad \boxed{z = cw}$$

Differentiation,

$$dx = a du, \quad dy = b dv, \quad dz = c dw.$$

eqⁿ ① becomes

$$V = \iiint a du \cdot b dv \cdot c dw.$$

$$V = abc \iiint u^0 v^0 w^0 du dv dw.$$

$$V = abc \iiint u^{1-1} v^{1-1} w^{1-1} du dv dw.$$

By Dirichlet's theorem,

$$V = abc \frac{\Gamma(1) \Gamma(1) \Gamma(1)}{\Gamma(1+1+1+1)}$$

$$V = abc \cdot \frac{0! 0! 0!}{\Gamma(4)} = \frac{abc (1)}{3!} = \frac{abc}{6}$$

$$\therefore \boxed{V = \frac{abc}{6}}$$

Unit 6, chapter-11

Centre of Gravity and Moment of Inertia.

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- * Centre of Gravity:- Centre of gravity (C.G.) is an imaginary point in a body of matter where, for convenience in certain calculations, that total weight of body may be thought to be concentrated.
- * Advantages of C.G. in Engineering:- The concept of C.G. is useful in designing static structures.
ex) building and bridges or in predicting the behaviour of a moving body when it is acted upon by gravity.
- * Note:- C.G. represent a point, That is in 2D it represents $(\bar{x}, \bar{y})$ and in 3D $(\bar{x}, \bar{y}, \bar{z})$.

* Types of C.G.

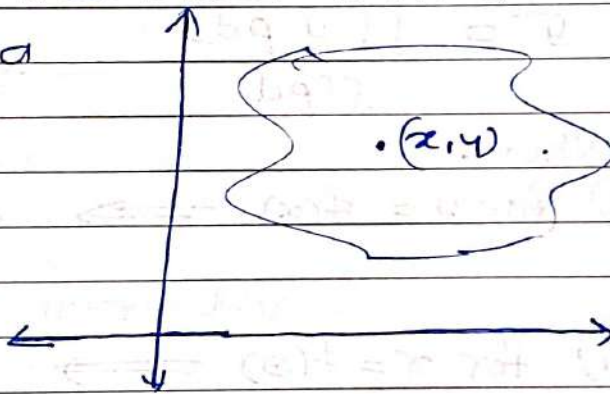
- ① C.G. of Plane Area of ~~a curve~~ Lamina.

The C.G. of plane area is

$$\bar{x} = \frac{\iint x \, dx \, dy}{\iint dx \, dy}$$

and

$$\bar{y} = \frac{\iint y \, dx \, dy}{\iint dx \, dy}$$



where ρ (density) is constant.

- * Note:- If density (ρ) is not constant then

$$\bar{x} = \frac{\iint \rho x \, dx \, dy}{\iint \rho \, dx \, dy} \quad \text{and} \quad \bar{y} = \frac{\iint \rho y \, dx \, dy}{\iint \rho \, dx \, dy}$$

② C.G. of Solids:-

$$\bar{x} = \frac{\iiint x \rho dx dy dz}{\iiint \rho dx dy dz}$$

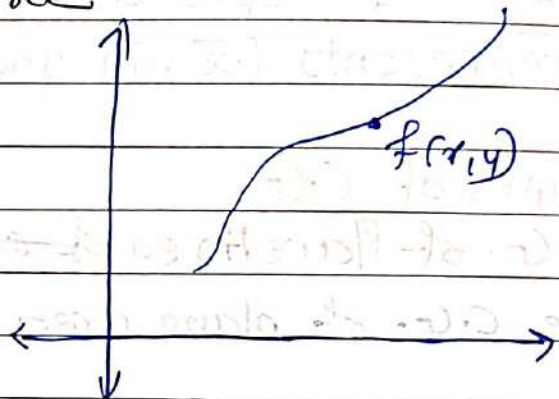
$$\bar{y} = \frac{\iiint y \rho dx dy dz}{\iiint \rho dx dy dz}$$

$$\bar{z} = \frac{\iiint z \rho dx dy dz}{\iiint \rho dx dy dz}$$

* C.G. of Arc of a Curve

$$\bar{x} = \frac{\iint x \rho ds}{\iint \rho ds}$$

$$\bar{y} = \frac{\iint y \rho ds}{\iint \rho ds}$$



where,

(i) for $y = f(x) \Rightarrow ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

(ii) for $x = f(y) \Rightarrow ds = \sqrt{r^2 + \left(\frac{dx}{dy}\right)^2} dy$

(iii) for $x = f_1(t)$ and $y = f_2(t) \Rightarrow ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

EX) Find the x co-ordinate of the C.G. of the area bounded by parabola $y^2 = x$ and the line $x + y = 2$.

$\Rightarrow$ step ① formula.

The x -coordinate of the C.G. of area is given by

$$\bar{x} = \frac{\iint x dx dy}{\iint dx dy} = \frac{M}{D}$$

step ② Diagram :-

$y^2 = x \Rightarrow$ Parabola, Symmetrical about x -axis
 $x + y = 2 \Rightarrow$ Straight line.

put $x = 0$

$$0 + y = 2$$

$$\boxed{y = 2}$$

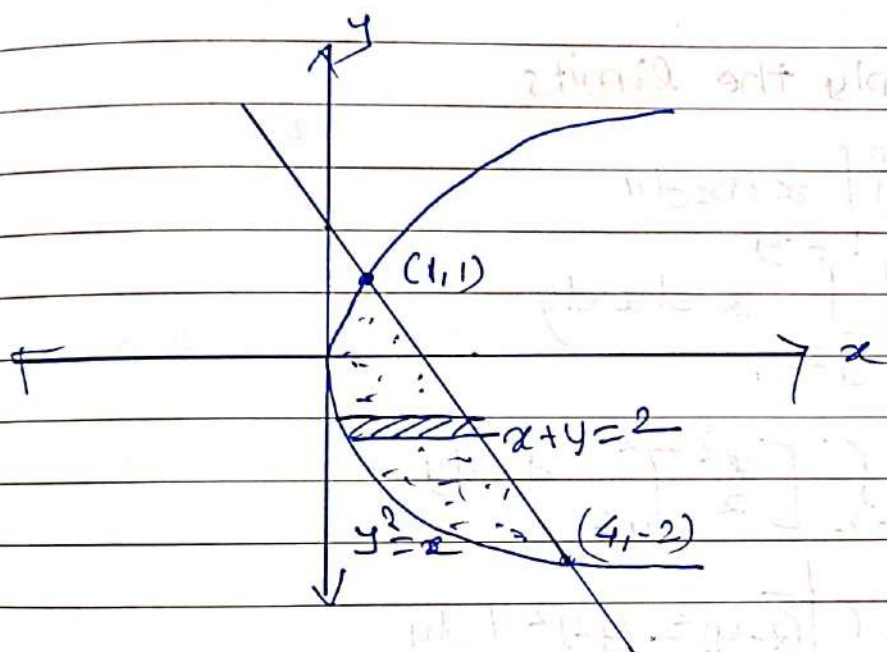
i.e. $(0, 2)$

put $y = 0$

$$x + 0 = 2$$

$$\boxed{x = 2}$$

i.e. $(2, 0)$



step ③ The points of intersection:-

$$\Rightarrow y^2 = x \text{ and } x + y = 2$$

$$\Rightarrow y^2 + y = 2$$

$$\Rightarrow y^2 + y - 2 = 0$$

$$y = 1 \text{ and } y = -2$$

for $y = 1$

$$x + 1 = 2$$

$$x = 2 - 1$$

$$\boxed{x = 1}$$

i.e. $(1, 1)$

and for $y = -2$

$$x - 2 = 2$$

$$x = 2 + 2$$

$$x = 4$$

i.e. $(4, -2)$

step ④: - ship

* Horizontal Ship *

Left end of the ship lies on Parabola

$$x = y^2$$

Right end of the ship lies on line

$$x + y = 2$$

$$\Rightarrow x = 2 - y$$

∴ To find limits of y , vary the ship from bottom to top.

y varies from $y = -2$ to $y = 1$.

* step ⑤ Apply the limits

$$N = \iint x \, dx \, dy$$

$$N = \int_{-2}^1 \int_{y^2}^{2-y} x \, dx \, dy$$

$$N = \int_{-2}^1 \left[\frac{x^2}{2} \right]_{y^2}^{2-y} dy$$

$$N = \frac{1}{2} \int_{-2}^1 [(2-y)^2 - (y^2)^2] dy$$

$$N = \frac{1}{2} \int_{-2}^1 (4 - 4y + y^2 - y^4) dy$$

$$N = \frac{1}{2} \left[4y - \frac{4y^2}{2} + \frac{y^3}{3} - \frac{y^5}{5} \right]_{-2}^1$$

$$N = \frac{1}{2} \left[\left(4(1) - 2(1)^2 + \frac{(1)^3}{3} - \frac{(1)^5}{5} \right) - \left(4(-2) - 2(-2)^2 + \frac{(-2)^3}{3} - \frac{(-2)^5}{5} \right) \right]$$

$$N = \frac{1}{2} \left[\left(4 - 2 + \frac{1}{3} - \frac{1}{5} \right) - \left(-8 - 8 - \frac{8}{3} + \frac{32}{5} \right) \right]$$

$$N = \frac{1}{2} \left[\left(\frac{60 - 30 + 5 - 3}{15} \right) - \left(\frac{-120 - 120 - 40 + 96}{15} \right) \right]$$

$$N = \frac{1}{2} \left(\frac{32}{15} - \left(\frac{-184}{15} \right) \right) = \boxed{\frac{36}{5}}$$

and

$$D = \iint dx dy = \int_{-2}^1 \int_{y^2}^{2-y} dx dy$$

$$D = \int_{-2}^1 [x]_{y^2}^{2-y} dy$$

$$D = \int_{-2}^1 (2 - y - y^2) dy$$

$$D = \left(2y - \frac{y^2}{2} - \frac{y^3}{3} \right)_{-2}^1$$

$$D = \left(2(1) - \frac{1^2}{2} - \frac{1^3}{3} \right) - \left(2(-2) - \frac{(-2)^2}{2} - \frac{(-2)^3}{3} \right)$$

$$D = \left(2 - \frac{1}{2} - \frac{1}{3} \right) - \left(-4 - 2 + \frac{8}{3} \right)$$

$$D = \left(\frac{12 - 3 - 2}{6} \right) - \left(\frac{-12 - 6 + 8}{3} \right)$$

$$D = \frac{7}{6} + \frac{10}{3} = \frac{7 + 20}{6} = \frac{27}{6} = \boxed{\frac{9}{2}}$$

$$\therefore \bar{x} = \frac{N}{D} = \frac{\frac{36}{5}}{\frac{9}{2}} = \frac{36}{5} \times \frac{2}{9} = \frac{8}{5}$$

$$\therefore \bar{x} = \frac{8}{5}$$

This is required $\bar{x}$ co-ordinate of G.G. of given area.

Que) Find the C.G of one loop of $r = a \sin 2\theta$.

⇒ step ① ∴

The curve $r = a \sin 2\theta$ consists of 4 loops.
consider the one loop in first quadrant
It is symmetrical about the line $\theta = \frac{\pi}{4}$ or $y = x$.

∴ Its C.G lies on the line $y = x$.

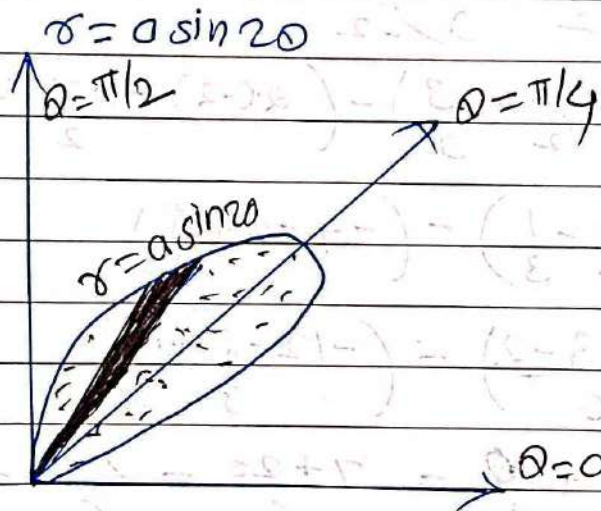
$$\therefore \bar{y} = \bar{x}.$$

Now

$$\bar{x} = \frac{\iint x \, dx \, dy}{\iint dx \, dy} \Rightarrow \frac{\iint r^2 \cos \theta \, dr \, d\theta}{\iint r \, dr \, d\theta} = \frac{N}{D}$$

(for polar coordinates)

step ② Diagram



step ③ Points of Intersection - Not Applicable.

step ④ Strip:-

slant strip

$$r = 0 \text{ to } r = a \sin 2\theta$$

$$\text{and } \theta = 0 \text{ to } \theta = \pi/2$$

step ⑤ Apply the limits

$$N = \int \int r^2 \cos \theta \, dr \, d\theta$$

$$N = \int_0^{\pi/2} \left(\int_0^{a(\sin \theta)} r^2 \, dr \right) d\theta$$

$$N = \int_0^{\pi/2} \left[\frac{r^3}{3} \right]_0^{a \sin \theta} \cos \theta \, d\theta$$

$$N = \frac{1}{3} \int_0^{\pi/2} \left[(a \sin \theta)^3 - 0^3 \right] \cos \theta \, d\theta$$

$$N = \frac{1}{3} \int_0^{\pi/2} \left[a^3 \sin^3 \theta \cdot \cos \theta \, d\theta \right]$$

but $\sin 2\theta = 2 \sin \theta \cos \theta$

$$\therefore N = \frac{a^3}{3} \int_0^{\pi/2} (2 \sin \theta \cos \theta)^3 \cos \theta \, d\theta$$

$$N = \frac{8a^3}{3} \int_0^{\pi/2} \sin^3 \theta \cos^4 \theta \, d\theta$$

By Reduction Formula,

$$N = \frac{8a^3}{3} \left[\frac{(3-1)(4-1)(4-3)}{7(7-2)(7-4)(7-6)} \times 1 \right]$$

$$N = \frac{8a^3}{3} \left(\frac{2 \times 3 \times 1}{7 \times 5 \times 3 \times 1} \times 1 \right) = \frac{16a^3}{105}$$

$$\therefore \boxed{N = \frac{16a^3}{105}}$$

and

$$D = \int \int r \, dr \, d\theta = \int_0^{\pi/2} \left(\int_0^{a(\sin \theta)} r \, dr \right) d\theta$$

$$D = \int_0^{\pi/2} \left(\frac{r^2}{2} \right)_0^{a \sin \theta} d\theta$$

$$\therefore D = \frac{1}{2} \int_0^{\pi/2} (a^2 \sin^2 2\theta - 0^2) d\theta$$

but $\sin 2x = 2 \sin x \cos x$

$$\therefore D = \frac{1}{2} \int_0^{\pi/2} a^2 \cdot 4 \sin^2 \theta \cos^2 \theta d\theta$$

$$D = \frac{4a^2}{2} \int_0^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta$$

By Reduction Formula,

$$D = 2a^2 \left[\frac{(2-1)(2-1)}{4(4-2)} \times \frac{\pi}{2} \right]$$

$$D = 2a^2 \left(\frac{\pi}{16} \right) = \boxed{\frac{\pi a^2}{8}}$$

$$\therefore \bar{X} = \frac{N}{D} = \frac{\frac{16a^3}{105}}{\frac{\pi a^2}{8}} = \frac{16a^3}{105} \times \frac{8}{\pi a^2} = \frac{128a}{105\pi}$$

$$\therefore \bar{x} = \frac{128a}{105\pi} \text{ and } \bar{y} = \frac{128a}{105\pi}$$

$\therefore$ C.G. of one loop of $r = a \sin 2\theta$ is

$$(\bar{x}, \bar{y}) = \left(\frac{128a}{105\pi}, \frac{128a}{105\pi} \right)$$

Que) Find the C.G. of the area of the cardioid

$$r = a(1 + \cos \theta)$$

$$\Rightarrow r = 0 \text{ to } r = a(1 + \cos \theta) \text{ and } N = \frac{5a^3\pi}{4}, D = \frac{3\pi a^2}{2}$$

$$\theta = 0 \text{ to } \theta = 2\pi$$

$$\bar{x} = \frac{5a}{6}, \bar{y} = 0$$

Que) Find the centroid of the loop of the curve
 $r^2 = a^2 \cos 2\theta$.

→ $r = 0$ to $r = a\sqrt{\cos 2\theta}$, $\theta = -\pi/4$ to $\pi/4$
 $N = \frac{a^3 \pi}{8\sqrt{2}}$, $D = \frac{a^2}{2}$, $(\bar{x}, \bar{y}) = (\frac{a\pi}{4\sqrt{2}}, 0)$.

* Moment of Inertia:-

A body's moment of Inertia is the product of its mass and square of the perpendicular distance of the mass from the axis of rotation. It is denoted by I .

$$I = mp^2$$

where $p =$ perpendicular distance (line).

* M.I. of an Area:-

$$I = \iint p^2 dx dy$$

or

$$I = \iint p^2 \rho dx dy \quad \text{where } \rho \text{ is density.}$$

$p =$ perpendicular distance

* M.I. of Solid:-

$$I = \iiint p^2 \rho dx dy dz$$

* M.I. of an Arc:-

$$I = \int p^2 ds \quad \text{where } ds = \frac{ds}{dx} \cdot dx.$$

or

$$ds = \frac{ds}{dy} \cdot dy$$

or

$$ds = \frac{ds}{dt} dt.$$

Que) Find the M.I. about the line $\theta = \frac{\pi}{2}$ of the area enclosed by $r = a(1 + \cos\theta)$

⇒ Step I:-

$$I = \iint r^2 dx dy$$

But $p = x$ (perpendicular to $\theta = \pi/2$ i.e. y-axis)

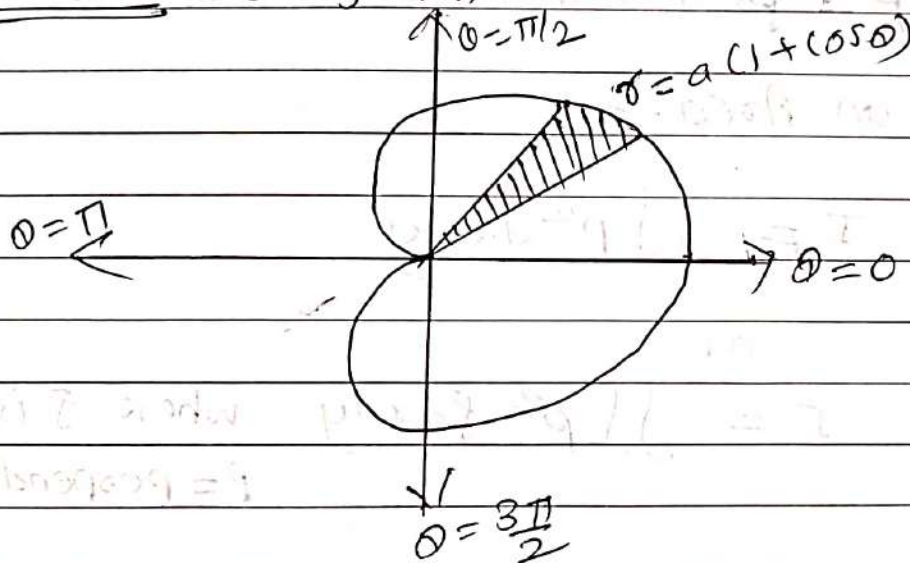
$$\Rightarrow I = \iint x^2 dx dy$$

converting to polar.

$$I = \iint r^2 \cos^2\theta (r dr d\theta)$$

$$I = \iint r^3 \cos^2\theta dr d\theta$$

* Step II:- Diagram



Step ③ Points of Intersection - Nil

Step ④ strip - slant strip.

$$r = a \text{ to } r = a(1 + \cos\theta)$$

$$\text{and } \theta = 0 \text{ to } \theta = \pi$$

(multiply by 2 to final answer for symmetry)

Step ⑤ Apply the limits.

$$I = \int_0^\pi \left(\int_a^{a(1+\cos\theta)} r^3 dr \right) \cos^2\theta d\theta$$

$$I = \int_0^\pi \left[\frac{x^4}{4} \right]_0^{a(1+\cos\theta)} \cos^2\theta d\theta$$

$$I = \int_0^\pi \left(\frac{a^4(1+\cos\theta)^4}{4} - \frac{0^4}{4} \right) \cos^2\theta d\theta$$

$$\therefore I = \frac{a^4}{4} \int_0^\pi (1+\cos\theta)^4 \cos^2\theta d\theta$$

$$I = \frac{a^4}{4} \int_0^\pi (1 + 4\cos\theta + 6\cos^2\theta + 4\cos^3\theta + \cos^4\theta) \cos^2\theta d\theta$$

$$I = \frac{a^4}{4} \int_0^\pi (\cos^2\theta + 4\cos^3\theta + 6\cos^4\theta + 4\cos^5\theta + \cos^6\theta) d\theta$$

but $\int_0^\pi \cos^n\theta = \int_0^{\pi/2} \cos^n\theta d\theta$ if n is even
 $= 0$ if n is odd

$$\therefore I = \frac{a^4}{4} \times 2 \int_0^{\pi/2} (\cos^2\theta + 0 + 6\cos^4\theta + 0 + \cos^6\theta) d\theta$$

$$I = \frac{a^4}{2} \left[\frac{(2-1) \cdot \pi}{2 \cdot 2} + \frac{6 \cdot (4-1)(4-3) \cdot \pi}{4(4-2) \cdot 2} + \frac{(6-1)(6-3)(6-5) \pi}{6(6-2)(6-4) \cdot 2} \right]$$

$$I = \frac{a^4}{2} \times \frac{\pi}{2} \left[\frac{1}{2} + \frac{3 \times 3 \times 1}{4 \times 2} + \frac{5 \times 3 \times 1}{2 \times 6 \times 4 \times 2} \right]$$

$$I = \frac{\pi a^4}{4} \left[\frac{1}{2} + \frac{9}{4} + \frac{5}{16} \right] = \frac{\pi a^4}{4} \left(\frac{8+36+5}{16} \right)$$

$$\boxed{I = \frac{49\pi a^4}{64}}$$

For symmetry $I = 2 \times \frac{49\pi a^4}{64} = \boxed{\frac{49\pi a^4}{32}}$ required answer.

Que) Find the moment of inertia of one loop of the lemniscate $r^2 = a^2 \cos 2\theta$ about initial line.

$\Rightarrow$ Let $\rho = \text{density}$
 $\therefore I = \iint r^2 \rho \, dx \, dy$

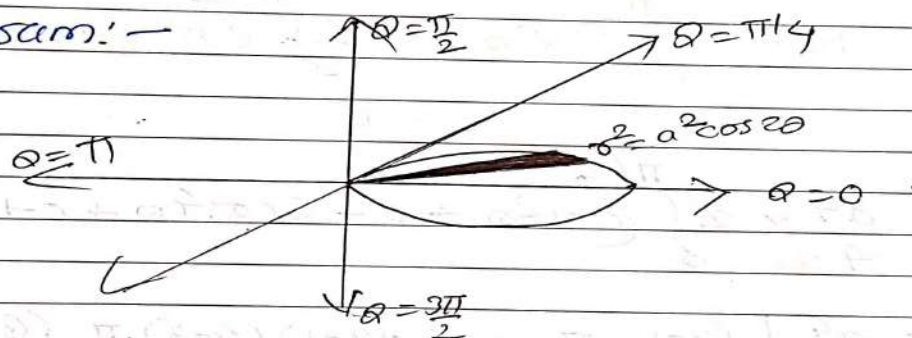
But $P = y$ (perpendicular to initial line)

$\therefore I = \iint y^2 \rho \, dx \, dy$
 converting to polar

$$I = \iint (r \sin \theta)^2 \rho \cdot (r \, dr \, d\theta)$$

$$I = \rho \iint r^3 \sin^2 \theta \, dr \, d\theta$$

* Diagram: -



* Strip: - slant strip

$$r = 0 \text{ to } r = a \sqrt{\cos 2\theta}$$

and

$$\theta = 0 \text{ to } \theta = \pi/4$$

(multiply by 2 to find answer for symmetry)

$\therefore$ Now $I = \rho \iint r^3 \sin^2 \theta \, dr \, d\theta$

$$I = \rho \int_0^{\pi/4} \left(\int_0^{a\sqrt{\cos 2\theta}} r^3 \, dr \right) \sin^2 \theta \, d\theta$$

$$I = \frac{8}{9} \int_0^{\pi/4} \left[\frac{x^4}{4} \right]_0^{a\sqrt{\cos 2\theta}} \sin^2 \theta d\theta.$$

$$I = \frac{8}{9} \int_0^{\pi/4} \left(\frac{a^4 \cos^2 2\theta}{4} - 0^4 \right) \sin^2 \theta d\theta.$$

$$I = \frac{8a^4}{9} \int_0^{\pi/4} (\cos^2 2\theta)^{1/2} \cdot \sin^2 \theta d\theta$$

$$I = \frac{8a^4}{9} \int_0^{\pi/4} \cos^2(2\theta) \sin^2 \theta d\theta.$$

(As $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$)

$$I = \frac{8a^4}{9} \int_0^{\pi/4} \cos^2(2\theta) \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

Substitute $2\theta = t \Rightarrow \theta = t/2$.

$2d\theta = dt \Rightarrow d\theta = dt/2$

When $\theta = 0 \Rightarrow t = 2(0) = 0$
 $\theta = \pi/4 \Rightarrow t = 2(\pi/4) = \pi/2$ } New limits.

$$I = \frac{8a^4}{9} \int_0^{\pi/2} \cos^2 t \left(\frac{1 - \cos t}{2} \right) \frac{dt}{2}$$

$$I = \frac{8a^4}{16} \int_0^{\pi/2} (\cos^2 t - \cos^3 t) dt.$$

$$I = \frac{8a^4}{16} \left[\frac{\cos t (2 - 1)}{2} \cdot \frac{\pi}{2} - \frac{(3 - 1)}{3 \cdot (3 - 2)} \cdot 1 \right]$$

$$I = \frac{8a^4}{16} \left(\frac{\pi}{4} - \frac{2}{3} \right) = \frac{8a^4}{16} \left(\frac{3\pi - 8}{12} \right)$$

$$\boxed{I = \frac{8a^4}{192}}$$

For symmetry

$$I = 2 \times \frac{\rho a^4 (3\pi - 8)}{96} = \frac{\rho a^4 (3\pi - 8)}{96}$$

Hence $\boxed{I = \frac{\rho a^4 (3\pi - 8)}{96}} \longrightarrow \textcircled{1}$

Now find the value of M to get ρ .

$$M = \rho \int_0^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} r dr d\theta$$

$$M = \rho \int_0^{\pi/4} \left[\frac{r^2}{2} \right]_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$M = \frac{\rho}{2} \int_0^{\pi/4} (a\sqrt{\cos 2\theta})^2 d\theta$$

$$M = \frac{\rho}{2} \int_0^{\pi/4} a^2 (\cos 2\theta) d\theta$$

$$M = \frac{\rho a^2}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{\pi/4}$$

$$M = \frac{\rho a^2}{4} [\sin(2(\frac{\pi}{4})) - \sin(0)]$$

$$M = \frac{\rho a^2}{4} (1 - 0)$$

For symmetry $M = 2 \times \frac{\rho a^2}{4} = \boxed{\frac{\rho a^2}{2}}$

As $M = \frac{\rho a^2}{2} \Rightarrow \boxed{\rho = \frac{2M}{a^2}}$

eqn (1) becomes

$$\therefore I = \frac{2M}{a^2} \cdot \frac{a^4 (3\pi - 8)}{96}$$

$$I = \frac{(3\pi - 8)ma^2}{48}$$

Find the moment of inertia of the portion of the parabola $y^2 = 4ax$, bounded by x -axis and latus rectum, about x -axis, if density at each point varies as cube of abscissa.

→ $I = \frac{12Ma^2}{11}$ (i.e. $\rho = kx^3$)
 where $M = \frac{4ka^5}{5}$.